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User-elicited dual-hand interactions for manipulating 3D objects in virtual reality environments

Vijayakumar Nanjappan¹, Hai-Ning Liang^{1*} , Feiyu Lu², Konstantinos Papangelis¹, Yong Yue¹ and Ka Lok Man^{1,3}

*Correspondence:

HaiNing.Liang@xjtlu.edu.cn

¹ Department of Computer Science and Software Engineering, Xi'an Jiaotong-Liverpool University, Suzhou, Jiangsu, China

Full list of author information is available at the end of the article

Abstract

Virtual reality technologies (VR) have advanced rapidly in the last few years. Prime examples include the Oculus RIFT and HTC Vive that are both head-worn/mounted displays (HMDs). VR HMDs enable a sense of immersion and allow enhanced natural interaction experiences with 3D objects. In this research we explore suitable interactions for manipulating 3D objects when users are wearing a VR HMD. In particular, this research focuses on a user-elicitation study to identify natural interactions for 3D manipulation using dual-hand controllers, which have become the standard input devices for VR HMDs. A user elicitation study requires potential users to provide interactions that are natural and intuitive based on given scenarios. The results of our study suggest that users prefer interactions that are based on shoulder motions (e.g., shoulder abduction and shoulder horizontal abduction) and elbow flexion movements. In addition, users seem to prefer one-hand interaction, and when two hands are required they prefer interactions that do not require simultaneous hand movements, but instead interactions that allow them to alternate between their hands. Results of our study are applicable to the design of dual-hand interactions with 3D objects in a variety of virtual reality environments.

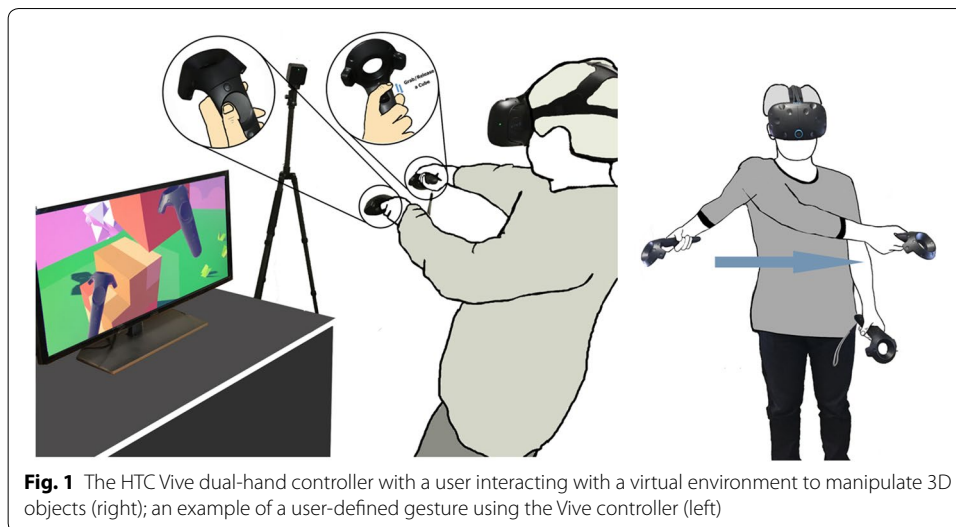
Keywords: Virtual reality, Head-mounted displays, Dual-hand input devices, Interaction design, 3D manipulation, Interaction techniques, User-elicitation study

Introduction

Virtual reality (VR) is still an emergent technology but it is rapidly advancing and marketed as a productivity and entertainment device. Examples include the HTC Vive¹ and Oculus RIFT² head-mounted displays (HMDs). Figure 1 below shows a picture of the HTC Vive and a scenario requiring a user to manipulate 3D virtual objects, one of the most common and basic activities in VR-based environments. There has been rapid progress in their displays in terms of portability and resolution and more recently there has been also a renewed focus on interaction (or input) devices. The HTC Vive for instance is the first to introduce its custom-designed dual-hand input controllers which

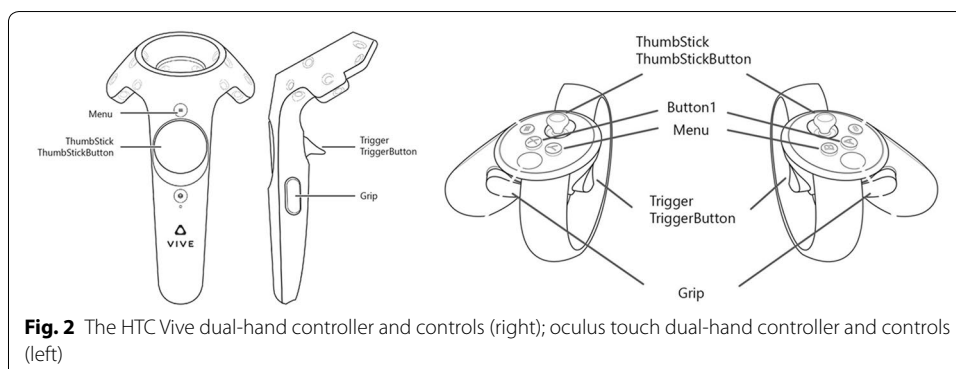
¹ <https://www.vive.com/us/product/vive-virtual-reality-system/>.

² <https://www.oculus.com/rift/>.



represent a significant departure from the traditional devices like the mouse + keyboard or game controller. The Oculus RIFT has followed soon after with the introduction of its own dual-hand controller, the Touch, which in many ways is very similar to the Vive controller. Figure 2 shows these two different dual-hand input devices. Further, recently-marketed mixed reality (MR) systems also come with these types of dual-hand devices. Typical examples include the Windows-based Lenovo and Samsung Odyssey MR HMDs. This trend in both VR and MR HMDs shows that dual-hand interaction is becoming standard across these systems.

One of the main benefits of VR is its immersive experience and the ability to manipulate and interact with 3D objects that approximates how they are manipulated in the physical world. The entire virtual world in the VR environments are collection of 3D objects. There are many ways to interact with 3D objects in these environments. One typical basic interaction is direct manipulation [1, 2] of 3D objects in these virtual worlds. In the real world, “manipulation” refers to any changes applied by/through the human hand. Similarly, in VR environments, a real hand or an interface tool is used to grasp and move 3D virtual artefacts. This virtual interaction enables a rich set of direct manipulations of these 3D objects. There are many fundamental forms of interaction in



a VR world: *moving around or navigation, selection, rotation, translation, scaling, slicing* and so on. These forms correspond to the actions we perform in the real world (e.g. navigation or rotation). There have been a great number of games and other types of applications coming out which require users to interact with 3D objects in a virtual environment. Minecraft for VR, a popular game that is supported by both the HTC Vive and Oculus, is a typical example of these types of games and requires users to use the dual-hand controllers to manipulate the 3D objects. Given their repaid introduction, further research is needed to know whether those forms of interaction based on dual-hand controllers are intuitive and preferable by typical users.

This research explores the suitability of dual-hand interactions for 3D manipulation in VR environments. Given that VR HMD have only become widely available, it is timely that we take a closer look at what types of interactions are considered natural and suitable for 3D manipulations by users. It is likewise important to know how people tend to perform manipulations when they are interacting with 3D objects inside a virtual reality environment. In this research, we conduct a study to elicit user-defined gestural interactions for dual-hand input devices for manipulating 3D virtual objects. This approach allows us to evaluate a series of tasks performed by users and extrapolate the set that is considered useful and intuitive by most of these users. This approach can also help extrapolate factors and rules that contribute to the design of these 3D manipulation techniques.

The main contributions of this paper are: (1) a set of user-elicited interactions for dual-hand manipulation of 3D objects in VR environments; and (2) recommendations for the design of these interactions.

Background and related work

This research is related to three themes: (1) content manipulation of 3D objects in VR environments, (2) dual-hand interaction, and (3) user elicited gestures.

Manipulating 3D objects in VR environments

Interaction with 3D objects is one of the most common activities in VR environments. There are two common implementations of VR environments: (1) spatially immersive projected wall displays (e.g. CAVEs), which show backside-projected computer-generated displays onto walls, ceilings, and floors of a small sized (often cubical) room [3]; and (2) head-worn/mounted displays (HMDs), which use binocular stereoscopic displays attached to users' head and have sensors to track and capture changes in the orientation of the users' head movements. These VR HMD devices support a variety of resolutions and are typically supported by 3D interfaces with different characteristics [4]. While the displays of VR goggles have advanced rapidly, input devices for these VR systems appear to be lagging behind. Different input/control devices have been investigated and evaluated before the arrival of dual-hand controllers, which are now becoming popular. Data

gloves are arguably the true hand gesture input device for direct manipulation of the 3D objects in virtual environments [5]. The traditional game controller for the Xbox and PlayStation systems are often used for interacting with 3D VR content—e.g., for playing games [6] and for navigation activities [7, 8].

Aside from the traditional game controllers, mobile devices have also been explored as input devices for interacting with 3D objects [9, 10]. The advantages of mobile devices include their portability and flexibility that allow customization and modification for a particular context [7, 11, 12]. Researchers have introduced several implementations based on the touch-enabled display and motion sensors of these mobile devices in order to turn them into controllers (e.g., see [11, 13–19]). It has also been shown that mobile devices could be useful for performing certain 3D tasks [15, 16, 18]. However, they are not frequently used to support activities in VR because of several limitations, such as the lack of tactile feedback and users not being able to see the touch display when they are wearing a VR HMD. Similarly, augmented reality (AR), serving as a technology that combines both real and virtual objects, requires the interaction with virtual objects to be quick, simple and intuitive. Such affordance can make the users more focused on the interactive contents themselves [20]. Device-free, mid-air interactions on 3D objects have also been evaluated by researchers in both AR and VR contexts [21, 22]. Le Chénéchal et al. [21] for example proposed a system to support hand-based manipulations in VR. Hand gestures are mapped and reconstructed to support collaboration activities. Bang et al. [23] have designed an interactive VR system that can be controlled by user postures. The Kinect is used to recognize human postures without the need for additional devices to be attached on the users' body.

We are now witnessing the introduction of a new generation of input controllers for VR HMDs. For example, the HTC Vive Controller, the Oculus RIFT Touch, and the PlayStation VR Move are all examples of these devices that allow dual-hand interactions. Given this development, it is timely that we investigate further dual-hand interaction with 3D objects in VR environments.

Dual hand vs. single hand input devices

Humans are proficient at using two hands but not for all types of tasks—some tasks can be proficiently done using both hands simultaneously, while others are better done by alternating their hands. Guiard [24] has proposed the Kinematic Chain Theory to describe the way people's hands work together to perform tasks in parallel. For bimanual tasks, it is often found that hands are assigned asymmetric roles, where one hand, usually the non-dominant one, determines the frame of references, while the other, dominant hand carries out more precise interactions within this frame. In some cases, researchers have also found that both hands can do symmetric roles with both hands performing similar actions [25].

Traditionally, the controller for gaming systems (such as the Xbox or PlayStation) let players conduct both asymmetrical actions (for example, the left thumb to control the left joystick while the right thumb set aside for pressing buttons) and symmetrical actions (with both thumbs controlling one joystick each). For mobile devices, research suggests that users prefer to use only one hand than two hands together [26]. Some studies have suggested that while two-handed interaction with mobile devices (e.g., for

typing and marking menus) can increase performance, the accuracy would get affected negatively in significant ways [25, 26]. Some efforts have also been made to develop interaction techniques for two-handed interactions using motion tracking devices [27]. However, it is shown that to be better than traditional input devices, they often require more training time and practice. This suggest that the design of such devices needs to be improved so that users could spend less time to get used to them.

In the context of VR HMDs, input devices like the Vive Controller or Oculus Touch are meant for dual hand interactions and it would appear that they are meant for synchronous, simultaneous interaction given that the pair of devices are physically exactly the same and have the same functionality. Given that they are relatively new, it is important to increase our understanding of how users would want to use them for 3D task manipulations in VR environments.

User-defined interaction gestures via elicitation

An elicitation technique is one type of data collection with the aim to help design interactions; it relies on potential users, rather than being based on the designers or developers themselves, as the source of inspiration [28]. The user elicitation approach is meant to guide the design of intuitive and natural interactions, as opposed to arbitrary designs that are technology- or designer-centric (e.g., to enhance the speed and accuracy of an algorithm in detecting spatial gestures). Earlier research by Nielsen et al. [29] has used elicitation to distil intuitive designs for ergonomic gestural interfaces. Besides being more intuitive and natural, user-defined gestures are easier to remember and are preferred by their potential non-technical users [12, 30–34]. Some researchers have reported that user-elicitation can actually help develop more complete sets of gestures than those defined only by experts or designers [34, 35]. There have been a number of researchers who have followed this approach to elicit users-defined gestures for mobile devices [33], tabletop systems [31, 34], public displays [36], skin input [37], cross-device interaction [38], and using tokens [39].

User-elicitation is particularly useful for devices which are new in the market and prototypes with which users have not had much experience. For example, in [18] a prototype of a mobile device based on two touch screens (one on the front and the other at the back) is introduced and a set of user-elicited gestures proposed for interacting with 3D content that is placed at a distance away from users. Similarly, in [40] the authors have explored the input space of smart glasses for gameplay. Our work falls into this category. This research deals with a technology that has only been available in the market for a short time: dual-hand controllers for VR HMDs. The two most common ones are the Vive controller and the Oculus Touch (see Fig. 2). The other distinguishing aspect is that we are interested in getting a deep insight into users' understanding and perception of this type of interaction with 3D VR content. With similar types of input devices being released for mixed reality HMDs, like the Samsung Odyssey, it is timely that we have a better understanding of how users feel about them and how they can intuitively use them to support their engagement with 3D content.

User-elicitation study

In order to find out what interactions are both natural and supportive for 3D manipulation in VR HMDs, we conducted an elicitation study. We wanted to conduct this study because we hoped to develop a more complete set of gestures and distil a set of design recommendations for such input devices. As stated earlier, user-elicitation can help to achieve this [34, 35, 40].

Apparatus and participants

Twelve right-handed non-paid participants (six females) with an average age of 21 were recruited for this experiment. They were all from a local university from different educational backgrounds. They had not had much first-hand experience with VR; 6 had seen how VR HMDs work in online videos. They all had normal or corrected-to-normal eye vision.

Our experimental prototype was based on the HTC Vive headset and its paired controller. Participants were given freedom to use any of the features of the controller, using one or both hands.

Task, procedure, and experimental design

Participants were asked to design and perform a gesture via the Vive Controller (a cause) to carry out the interaction (the effect). There were 17 tasks that had been identified based on reviewing relevant literature on 3D task manipulation (see Table 1). These tasks correspond to the fundamental forms of direct manipulation tasks in any kind of VR environments. We asked participants to do a gesture twice and explain why they chose a particular way of enacting the interaction. Before the experiment, participants were given a brief description of the features and interaction possibilities of the Vive Controller and asked to perform some movements they felt natural, intuitive, and comfortable with the device. The participants were asked to be standing during the experiment and were allowed to rest at any time if they feel tired during the experiment.

Participants were first given time to practice with the Vive Controller and HMD headset. To give them some focus, we let them play an in-house made game in which they had to manipulate a set of rectangular objects (see Fig. 1). It was also intended to help participants familiarize themselves with a typical 3D VR environment. The participants were asked to use the Vive controller to select and move the rectangular objects. After they were familiar with the virtual environment and the Vive controllers and headset, we began the experiment. We first showed the participants each interaction one at the time via 3D animations in the VR environment. After the animation was run once, a researcher would explain the 3D task further for clarity and asked if there were any questions. The animation could be replayed as many times as requested. Participants were then asked to create an interaction with the Vive Controller using any of the embedded features and in any manner that they felt intuitive. While performing the interaction, participants were asked to think aloud—that is, to verbalize what they were doing and why. Afterwards, they were asked to sketch

Table 1 The 3D tasks given to participants organized by category

3D manipulation tasks	
Manipulation	Animation descriptions
<i>Selection</i>	
1. Selection	Use Vive controller to <i>Select</i> a Cube
<i>Translation</i>	
2. About X axis	Move the cube along the <i>X-axis</i> : the axis pointing horizontally to the left or right
3. About Y axis	Move the cube along the <i>Y-axis</i> : the axis pointing vertically upward or downward
4. About Z axis	Move the cube along the <i>Z-axis</i> : the axis pointing forward or backward
5. XZ plane	Move the cube in the <i>X–Z plane</i>
6. YZ plane	Move the cube in the <i>Y–Z plane</i>
7. XY plane	Move the cube in the <i>X–Y plane</i>
<i>Rotation</i>	
8. About X axis	Rotate the cube <i>about the X axis</i>
9. About Y axis	Rotate the cube <i>about the Y axis</i>
10. About Z axis	Rotate the cube <i>about the Z axis</i>
<i>Throwing</i>	
11. Forward	Throw the cube <i>forward</i>
12. Backward	Throw the cube <i>backward</i>
13. Left	Throw the cube to the <i>left</i> side
14. Right	Throw the cube to the <i>right</i> side
15. Upward	Throw the cube <i>upward</i>
<i>Switch and stack</i>	
16. Switching	Switch the cube from <i>one hand</i> to <i>another</i>
17. Stacking	Stack one cube <i>on top</i> of another

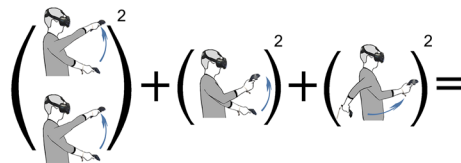
or write a brief description of the interaction on a piece of paper. The process was repeated for all 17 manipulation interactions shown in Table 1.

Results

From the collected gestures, we were able to create a set that were natural to users. We grouped identical gestures for each task, and the largest group was chosen as the user-defined gesture for the task. We then calculated an agreement score [34, 41] for each task using the group size. The score reflected in one number the degree of consensus among participants. The formula for calculating the agreement score is shown below

$$A_t = \sum_{P_i} \left(\frac{P_i}{P_t} \right)^2$$

where t is a task in the set for all tasks T ; P_t is the set of proposed gestures for t ; and P_i is a subset of identical gestures from P_t . The range for A_t is between 0 and 1 inclusive. As an example, let us assume that for a task, four participants gave each a gesture, but only two are very similar. Then the agreement score calculated following the process shown in Fig. 3.



$$A_t = \left(\frac{2}{4}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2 = 0.375$$

Fig. 3 An example of calculating the agreement score for a user elicited gesture

Figure 4 shows the agreement scores for the gesture set, ordered in descending order. The scores show consistent high agreement of participants based on the tasks. Figures 5, 6, 7, 8, and 9 show all the user-defined gestures based on category.

In this section we present the general observations of the study and then present some design considerations derived from the results.

From Fig. 6, we can observe that, in broad terms, there is high consistency for tasks requiring translation, especially along one axis only. There is lower agreement for

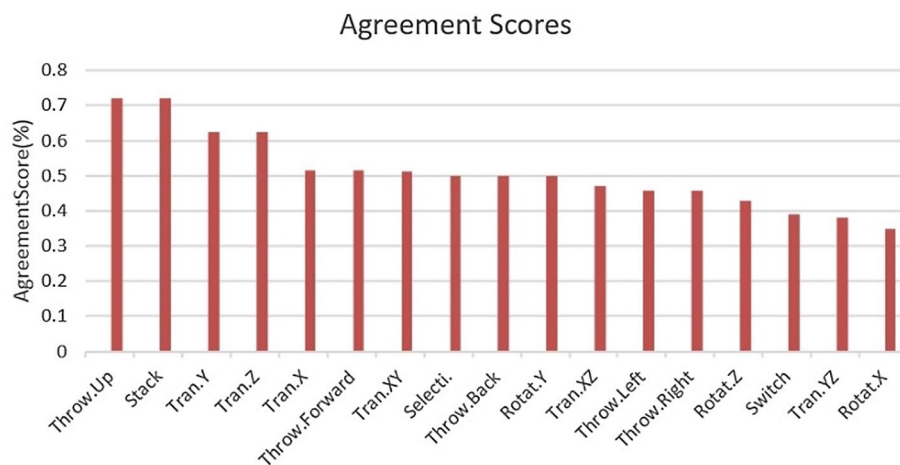


Fig. 4 The agreement scores for the tasks arranged in descending order

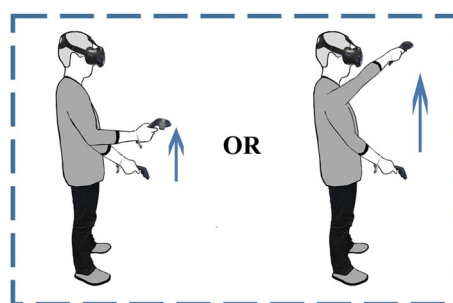
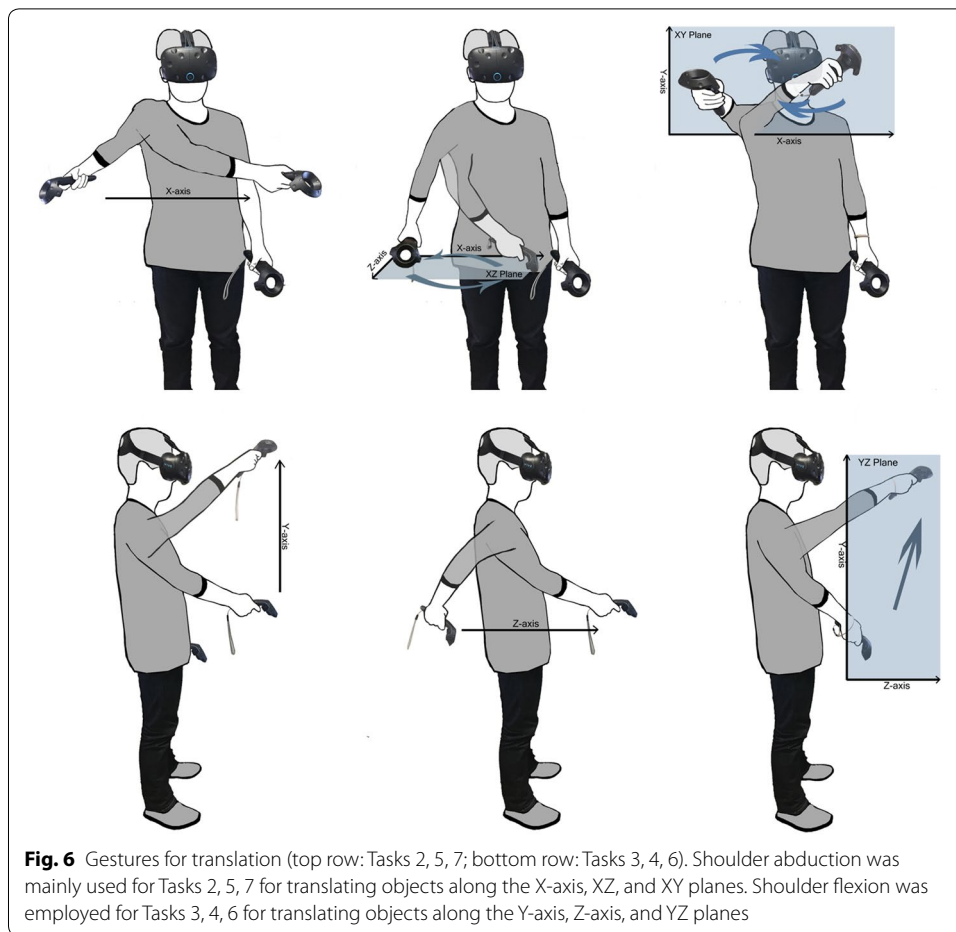


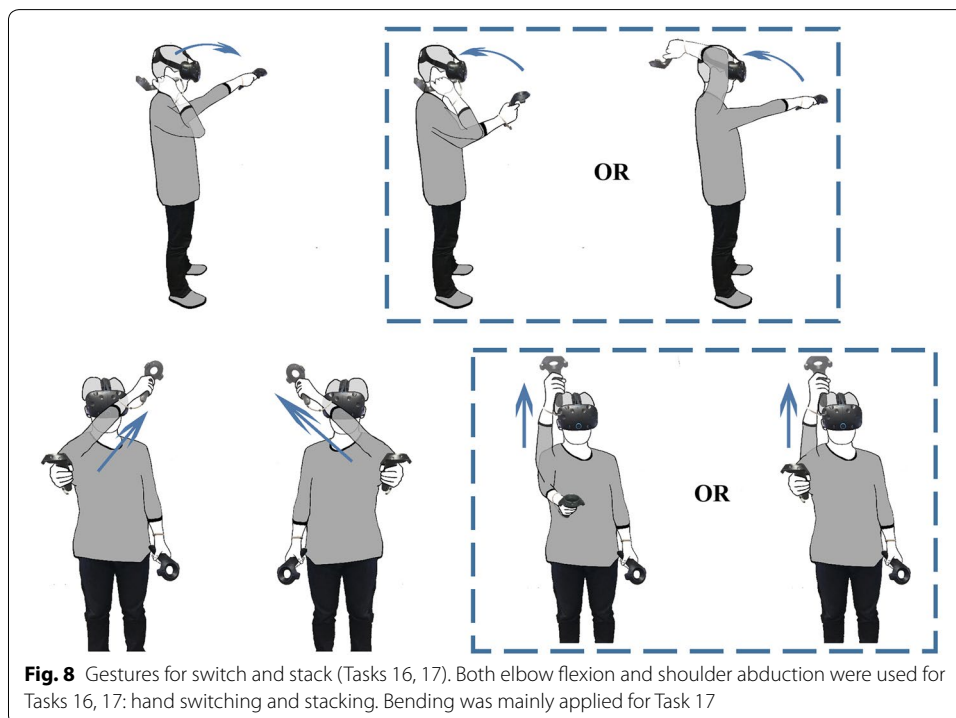
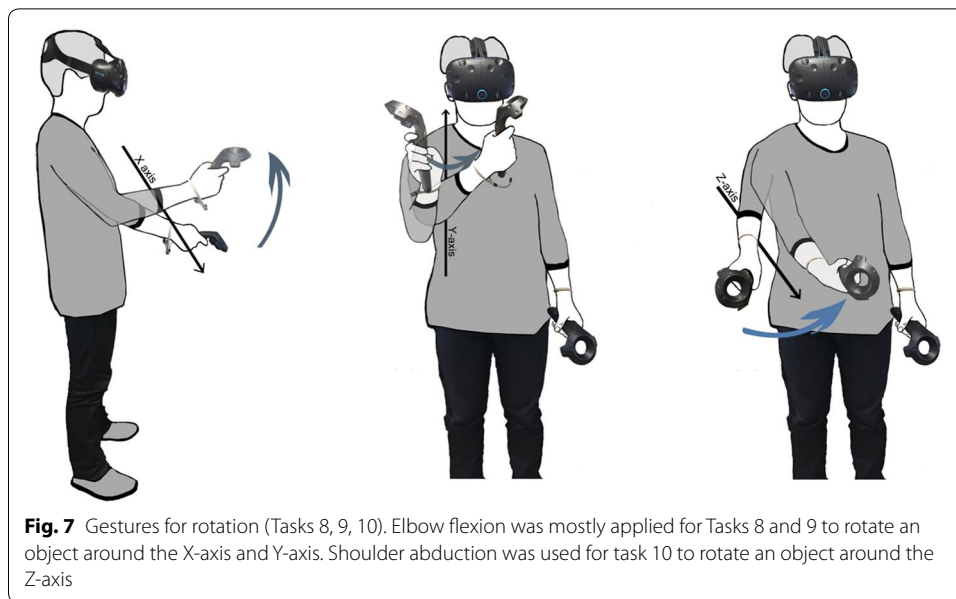
Fig. 5 Gestures for selection (Task1). Participants gave two ways for doing this task; left: elbow flexion; right: shoulder flexion



translation tasks dealing with the Z-axis (e.g., translation along the XZ axes). This observation agrees with results from [18] whose authors have examined user-elicited interactions for tasks using a mobile device and found that 3D tasks dealing with the Z-axis are not easy to perform and that users would give a wider range of possible interactions.

Another important pattern we observed is that most users liked doing large gestures with their arms. This is aligned with studies by [42, 43], where participants seem to have used stretch-out arm motions to select targets around their body. Table 2 shows the parts of the arms used to perform the category of gestures for the given 17 tasks by the participants. To understand why and how these patterns have emerged, we would need to look at the possible range of motions afforded by the human arm (see Fig. 10).

The human arm has three joints: shoulder, elbow, and wrist. They connect the arm proper, forearm, and hand. The shoulder and elbow joints allow people to carry out 5 distinct types of motions (see Fig. 10), while the wrist joint allows us to perform 3 types of rotations [44, 45]. Our study shows that participants have not made use of wrist motions but instead they have used either shoulder and elbow movements, or combinations of the two. That is, it is based on shoulder abduction, elbow flexion, shoulder internal rotation, shoulder flexion, and shoulder horizontal abduction.



In the exit questionnaire and interview, we asked participants why they had not made much use of their wrist. The common response was that given that the more natural way to work with the Vive Controller was to be standing, it was then also easier and more practical to use arm movements. In addition, they indicated that wrist motions were more suitable for small, minute movements that required precision and accurate control. They also commented that wrist rotations/twists were limited and physically difficult to

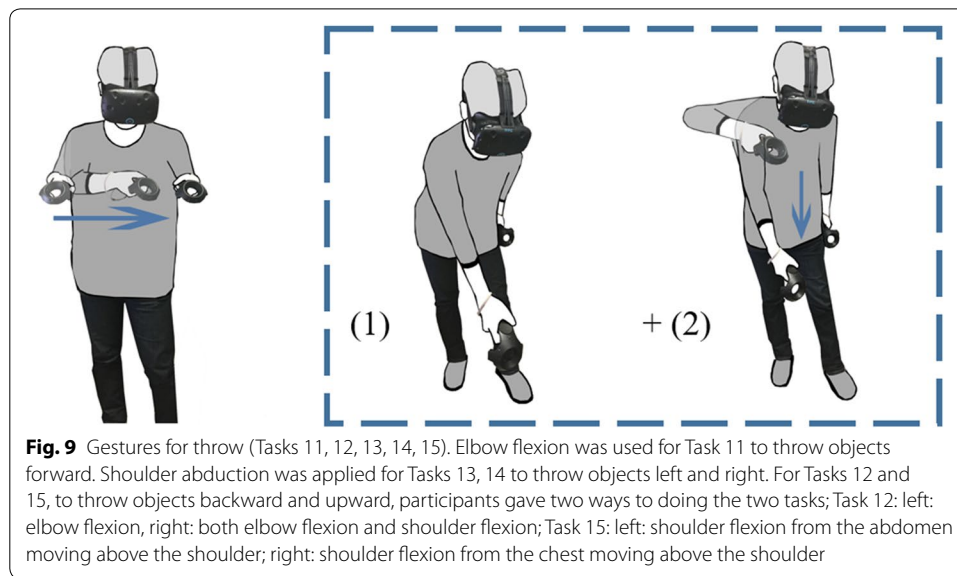
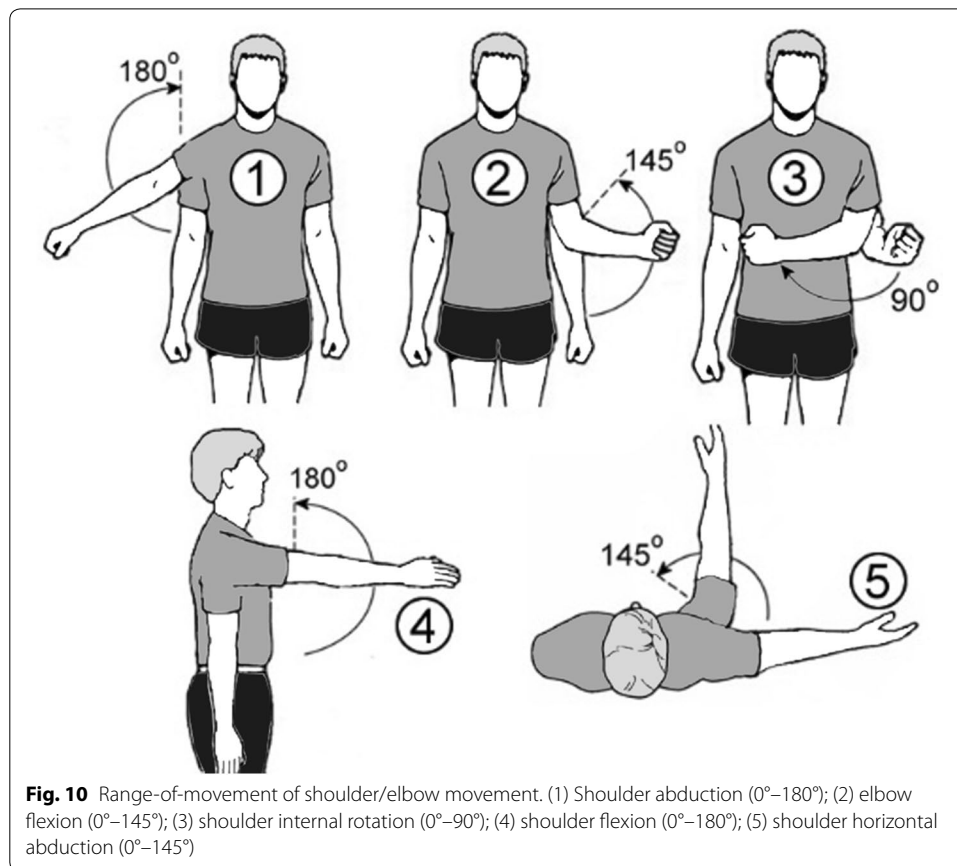


Table 2 Types of gestures performed by the participants based on the part of arm used

Part of the arm	Abduction	Flexion
Shoulder	Selection (Task 1) Translation (Tasks 2, 5, 7) Rotation (Task 10) Switch and stack (16 and 17)	
Elbow		Selection (Task 1) Translation (Tasks 3, 4, 6) Rotation (Tasks 8 and 9) Switch and stack (16 and 17)

do. Such comments were in line with research that had dealt with using the wrist for doing tilt gestures (e.g., see [45]), which had pointed out that the range of motions for human wrist was rather limited. Furthermore, participants also felt that VR should not be for very accurate tasks and that they would prefer to use a different controller for such tasks. One reason was that viewing through the HMD to focus on small elements for a prolonged time was tiring for their eyes; the other reason was that performing interactions that required precision in mid-air was not that easy (but was actually tiring). Although the touchpad located on top of each device could be used for accurate interactions, participants felt that it was not natural and convenient to use. Also, they suggested that the touchpad did not fit into their model of how the handheld device ought to work. They also indicated that it was not easy to interact with the touchpad while also carrying out motion gestures with the device. This seemed to agree with earlier research about input devices, where it was found that despite providing multiple features that were considered useful, actual users may not be able to take advantage of them and use more than one feature simultaneously [18, 46].

In terms of the 5 shoulder/elbow motions, participants seem to prefer shoulder motions (e.g., Task 1-right, 2, 3, 4, 5, 6, 7, 10, 13, 14, 15), followed by elbow ones (e.g., Tasks 1-left, 8, 9, 11, Task 12-left). There were some cases when both shoulder and elbow



were used together (e.g. 12-right, 16, 17) and at least one gesture which required bending (e.g., 17). In a way, most of the gestures appear to be a combination shoulder abduction, flexion, and horizontal abduction. We asked participants to provide their preferences for each type of shoulder and elbow motion. Participants' responses were to a large extent consistent in this order: Shoulder flexion → shoulder abduction → elbow flexion → shoulder horizontal abduction → shoulder internal rotation. When asked why, most of them said that they had more flexibility of motion and felt less tired with the first three motions, while the last two would increase their fatigue level but provided them with small range of motions. Support for this can be found in literature for gestural interactions. For example, techniques proposed by [42, 43] rely on shoulder flexion accompanied by small shoulder horizontal abduction movements; and techniques recommended in [47] for menu selection of self-portrait cameras are based on shoulder flexion plus elbow flexion with small shoulder horizontal motions.

In terms of using the two hand-held devices together, it would appear that for most tasks participants were able to do the tasks with one of the devices. In addition, participants commented that the two devices were identical and sometimes was difficult to tell them apart and because of this it was not easy to think in terms of using the two together. In terms of its design, participants said that it would have been more cognitively easier if the two devices had a different feeling, tactile wise or in terms of its shape, something akin to the Nintendo Wii Remote and the Nunchuk combination. Despite the

form factor, participants were observed using two devices for coordinated tasks, such as Task 16, which was primarily based on elbow flexion/internal rotation. We asked participants if they would want to use two hands more often. They said that they prefer not to because it is not easy to coordinate two hands moving together, especially when they were wearing the VR HMD. This was rather surprising. They said that if they had to use two hands, they would not mind doing so but they would rather use one hand. If they had to use two hands, a number of participants suggested it would be preferable if the two hands were going in the same direction or doing the same activity—for example, both hands doing elbow flexion. In addition, they said that they would find it difficult if two hands were doing things simultaneously. They further said that it would be better if interactions requiring two hands to have asynchronous actions, with one hand doing one action first, and afterward the other hand can perform a follow up action. This is in line with research about dual hand interaction techniques [20, 25, 26, 48, 49]. For example, for text entry activities, when using two hands at the same time, it can lead to faster performance, but it can also decrease their accuracy [26, 48]. In addition, when performing two-handed simultaneous marking menu strokes, it has been reported that the participants had the slowest reaction time because of the extra cognitive burden in “remembering and planning their strokes when coordinating simultaneous motions of two hands” [42; p. 16.14].

Discussion and conclusions

Lessons learned

The following design guidelines are based on the lessons gathered from our results.

- 3D manipulations with a handheld device should attempt to minimize the use of the wrist.
- VR environments should minimize requiring users to carry out precise actions that require focusing on small elements for a long time.
- Interactions should try to leverage shoulder flexion and shoulder abduction in combination with elbow flexion.
- Stretching-out and lifting of the arms appears to be more preferred and easier to perform.
- Simultaneous dual hand interactions seem suitable for tasks that may not require precise movements.

Future work

As the proliferation of dual-hand controllers for both virtual and mixed reality systems continues to grow, further research will still be needed as they are still open questions. The dual-hand controllers for VR systems are symmetrical and have the same functionalities. We have the example of the Nintendo Wii controller, which in a way is a combination of two separate input devices (the Wii Remote and the Nunchuk). Although it is a dual-hand controller, each side has a different form factor (for example their shapes and tactile feelings are different) and yet have both similar and different functions. It would be a useful line of research to explore if similar asymmetric designs will enhance the usability of the

controllers for manipulating 3D objects in VR environments. All current dual-hand VR controllers appear to have symmetrical design and functionalities for both hands. With an asymmetrical design, it will be possible to explore how one can leverage Guiard's theory of Kinematic Chain [50] in VR environments. For example, we can answer questions like how to use one hand to serve as the reference frame to support and complement the other simultaneous, more precise tasks that need to be performed with the other hand; or whether asymmetrical controllers can lead to better synchronization of simultaneous activities using both hands at the same time in VR environments.

In addition, manipulating 3D objects in VR environments can be affected by the nature and properties of the objects themselves. This effect can be multi-faceted and these factors include size of the objects, their distance from the users, their shape (e.g., regular vs. irregular), and whether the objects are moving or static. To identify any correlations between these properties of virtual 3D objects and users' preferred choice of gestures requires further investigation, especially when these gestures are based on dual hand devices.

Summary and conclusions

In the paper we have presented our work on the exploration of suitable manipulations to interact with 3D objects in virtual reality head-mounted display (HMD) environments. We conducted a user-elicitation study to explore what interactions are more natural and intuitive for dual-hand controllers for manipulating 3D objects in these environments. The results of the study suggest that for dual-hand devices users prefer interactions that are based on shoulder motions (e.g., shoulder abduction, shoulder horizontal abduction) and elbow flexion movements. In addition, users seem to prefer one-hand interaction, and when two hands are required they prefer interaction that do not require simultaneous hand or arm movements for precise interactions. Our research is limited to one type of dual-hand controller (based on the HTC Vive). Despite this, our results are applicable to similar dual hand devices, like the Oculus Touch and PlayStation Move. In the future we plan to explore design issues of these dual-hand controllers to see if we can increase users' preference and ability to use two hands to interact with 3D objects in virtual environments.

Authors' contributions

VN, H-NL, and FL framed the ideas of the paper and the design of the study. VN and FL run the experiment to collect the data and, with the help of H-NL, did the analysis of the results. All 6 authors (VN, H-NL, FL, KP, YY and KLM) contributed to the writing of the paper. All authors read and approved the final manuscript.

Author details

¹ Department of Computer Science and Software Engineering, Xi'an Jiaotong-Liverpool University, Suzhou, Jiangsu, China. ² Department of Computer Science, Virginia Tech, Blacksburg, VA, USA. ³ Faculty of Engineering, Computing, and Science, Swinburne University of Technology Sarawak, Sarawak, Malaysia.

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Competing interests

The authors declare that they have no competing interests.

Availability of data and materials

Available upon request.

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